Muhammad Talha Atif

Backend Engineer

+923064391520 ◇ talhaatif573@gmail.com ◇ Lahore, Pakistan ◇ LinkedIn ◇ GitHub ◇ Portfolio

SUMMARY

Backend Engineer with 1+ year of production experience building high-performance, distributed backend systems using Java, Spring Boot, and Node.js. Designed and delivered mission-critical microservices with Kafka event-driven pipelines, Redis caching, and PostgreSQL at enterprise scale. Deep understanding of data structures, algorithms, and distributed system architecture through production work and 500+ competitive programming problems solved. Experienced in high-availability service design, real-time event processing, and cross-functional collaboration on rapidly iterating products.

WORK EXPERIENCE

Backend Engineer *Dec '25 – Present*
MicroTechX Remote

Designed and maintained high-availability Java backend services for an enterprise IAM platform, building distributed microservices with event-driven processing, real-time state tracking, and scalable API architecture across 8+ services.

- Architected Kafka event-driven microservices for real-time audit trails and approval workflows across 8+ distributed Java services, tuning consumer concurrency and dead-letter queues for high-throughput fault-tolerant processing.
- Designed a high-performance bulk assignment API processing 5,000+ records per operation using batch queries and optimised PostgreSQL writes, eliminating synchronous bottlenecks and keeping latency flat under load.
- Led zero-downtime OpenSearch migration reindexing 5M+ documents via alias switching, resolving mapping conflicts and migrating high-cardinality fields to improve aggregation query performance across the analytics layer.

Backend Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Built and iterated production backend APIs for a high-traffic social platform, optimising system performance, real-time reliability, and data consistency under concurrent user load.

- Optimised high-traffic feed API using selective database indexing, DTO pruning, query rewriting, and a Redis in-memory cache, reducing response time from 3 seconds to 700ms under real user load.
- Resolved a concurrency bug causing notification duplication by replacing non-transactional writes with database transactions, eliminating race conditions and restoring data consistency across the notification system.
- Implemented IP-based rate limiting across authentication endpoints and deployed a zero-downtime Docker migration to Fly.io, maintaining service availability throughout the infrastructure transition.

PROJECTS

Moodly | AI Experience Booking Platform | Demo

- Built Python inference service classifying emotions across 7 classes from real-time voice and facial input, with RabbitMQ async pipeline decoupling inference from booking under concurrent load.
- Designed high-availability booking lifecycle APIs with request validation, conflict prevention, and unit-tested state transitions across all critical paths.

Memingo | Language Learning Platform | Demo

- Resolved a critical read-path bottleneck by replacing per-request ranking computation with a Redis precomputed cache, reducing leaderboard API latency from 30 seconds to 90ms.
- Built real-time object detection module returning structured vocabulary responses as a single low-latency API call, integrated with an AI chatbot with session-aware context management across learning sessions.

SKILLS

Languages Java, Python, TypeScript, JavaScript, SQL, Kotlin, Go (learning)

Backend Spring Boot, Spring MVC, Spring Security, Node.js, Express.js, NestJS, FastAPI, REST APIs, gRPC
Distributed Systems Kafka, RabbitMQ, event-driven architecture, microservices, WebSockets, real-time processing
Databases PostgreSQL, MySQL, MongoDB, Redis, SQLite, Supabase, Firebase, Hibernate/JPA, OpenSearch
System Design High availability, distributed architecture, caching strategies, batch processing, fault tolerance
Security JWT, OAuth 2.0, RBAC, rate limiting, Spring Security, input validation
Cloud / DevOps AWS (S3, RDS, EC2), Docker, Kubernetes (familiar), GitHub Actions, CI/CD, Maven
Tools Postman, Swagger, Stripe, VS Code, IntelliJ IDEA

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

Participant, CALICO Berkeley Speed Coding Competition (2024) *Dec '24*
University of California, Berkeley

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*